

Postulate 3: (commutative) (a) $x+y=y+x$ (b) $x \cdot y=y \cdot x$
 Postulate 4: (distributive) (a) $x(y+z)=xy+xz$ (b) $x+yz=(x+y)(x+z)$

2.3.2 Basic Theorems

Theorem 1(a): $x+x=x$

Proof: $x+x = x+(x \cdot 1)$ (Postulate 2(b))
 $= x+(x \cdot (x+x'))$ (Postulate 5(a))
 $= x+(x \cdot x + x \cdot x')$ (Postulate 4(a))

(Wrong approach)

$x+x = (x+x) \cdot 1$ (Postulate 2(b))
 $= (x+x) \cdot (x+x')$ (Postulate 5(a))
 $= x+x \cdot x'$ (Postulate 4(b))
 $= x+0$ (Postulate 5(b)) $= x$ (Postulate 2(a)) $\square$

→ Fantastic

Theorem 1(b): $x \cdot x = x$

Proof: $x \cdot x = x \cdot (x+x)$ (Theorem 1(a))
 $= x \cdot x + x \cdot x$

(Wrong approach)

$x \cdot x = x \cdot (x+0)$ (Postulate 2(a))
 $= (x+0) \cdot (x+0)$ (Postulate 2(a))
 $= x+0 \cdot 0$ (Postulate

(Wrong approach)

$x \cdot x = (x \cdot x) + 0$ (Postulate 2(a))
 $= (x \cdot x) + x \cdot x'$ (Postulate 5(b))
 $= x \cdot (x+x')$ (Postulate 4(a))
 $= x \cdot 1$ (Postulate 5(a))
 $= x$ (Postulate 2(b))
 (Fantastic) $\square$

Theorem 2(a): $x+1=1$

Proof: $x+1 = (x+1) \cdot 1$ (Postulate 2(b))
 $= (x+1) \cdot (x+x')$ (Postulate 5(a))
 $= x+1 \cdot x'$ (Postulate 4(b))
 $= x+x' \cdot 1$ (Postulate 3(b))
 $= x+x'$ (Postulate 2(b))
 $= 1$ (Postulate 5(a)) $\square$